Binary-Model Applications and Final Review – Supplement

18 Nov 2026 | Chapter 11 and cumulative review

ECO 351 – Introduction to Econometrics

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Review: inference and identification

A statistically significant coefficient concerns sampling evidence under the maintained model.

A confidence interval quantifies uncertainty about the specified estimand under its assumptions.

A causal interpretation requires an argument that the comparison isolates the effect of interest. Neither a small p-value nor a high R^2 supplies that argument.

In your revision, practise stating the null, decision, magnitude, and identifying assumption separately.

Supplement: the indicator-model coefficient output

```
. logit deny c.pir i.black, vce(robust)
Logistic regression                                     Number of obs =  2,380
------------------------------------------------------------------------------
             |               Robust
        deny | Coefficient  std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         pir |   5.370362   .9633435     5.57   0.000     3.482244    7.258481
     1.black |   1.272782   .1460986     8.71   0.000      .9864339     1.55913
       _cons |  -4.125558    .345825   -11.93   0.000    -4.803362   -3.447753
------------------------------------------------------------------------------
```

Supplement: local marginal effects

```
. margins, dydx(pir) at(pir=(0.30 0.40))
Conditional marginal effects                                    Number of obs = 2,380
------------------------------------------------------------------------------
             |            Delta-method
             |      dy/dx   std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
pir          |
         _at |
          1  |   .5061048   .0687362     7.36   0.000      .3713842    .6408254
          2  |   .7131224   .1318454     5.41   0.000      .4547102    .9715347
------------------------------------------------------------------------------
```